

Assignment 7

Q1. Create EC2 Instance and Connect

Create a basic virtual machine.

Tasks:

Go to Amazon EC2 Dashboard

Launch instance:

Name:

basic-ec2

AMI Ubuntu

Type:

t2.micro

Create a new key pair and download it

Create Security Group:

Allow SSH (port 22) from your IP

Launch instance

Connect using EC2 Instance Connect (browser)

The screenshot displays the AWS Management Console interface for an EC2 instance. At the top, the 'Instances (1/1)' section shows a table with one instance, 'assign-ec2', which is in the 'Running' state. Below this, the 'Details' tab for the instance 'i-0d2de7c6cf0c00290' is selected. The details are organized into three columns: General information (Instance ID, IPv6 address, Hostname type, Answer private resource DNS name, Auto-assigned IP address, IAM role), Networking (Public IPv4 address, Private IPv4 addresses, Public DNS, Private DNS name, Instance type, VPC ID, Subnet ID), and Other (Elastic IP addresses, AWS Compute Optimizer finding, Auto Scaling Group name).

Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
assign-ec2	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1c	ec2-98-93-99-176.com...	98.93.99.176	-

i-0d2de7c6cf0c00290 (assign-ec2)						
Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
Instance summary Instance ID: i-0d2de7c6cf0c00290 IPv6 address: - Hostname type: IP name: ip-172-31-21-120.ec2.internal Answer private resource DNS name: IPv4 (A) Auto-assigned IP address: 98.93.99.176 [Public IP] IAM role: -	Public IPv4 address : 98.93.99.176 open address Instance state : Running Private IP DNS name (IPv4 only) : ip-172-31-21-120.ec2.internal Instance type : t3.micro VPC ID : vpc-01b92e3d8449ea700 (default-don't delete) Subnet ID : subnet-07a1c13caa24971cf (def3)	Private IPv4 addresses : 172.31.21.120 Public DNS : ec2-98-93-99-176.compute-1.amazonaws.com open address Elastic IP addresses : - AWS Compute Optimizer finding : Opt-in to AWS Compute Optimizer for recommendations. Learn more Auto Scaling Group name : -				

```
ubuntu@ip-172-31-21-120: ~  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Mon Apr 27 09:29:19 UTC 2026  
  
System load:  0.03      Temperature:   -273.1 C  
Usage of /:   26.9% of 6.71GB  Processes:    113  
Memory usage: 25%      Users logged in: 0  
Swap usage:   0%        IPv4 address for ens5: 172.31.21.120  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Mon Apr 27 09:25:10 2026 from 18.206.107.28  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@ip-172-31-21-120:~$ |
```

Q2. Create S3 Bucket and Upload File

Work with storage using GUI.

Tasks:

Go to Amazon S3

Create bucket:

Name:

my-basic-bucket-001

(must be unique)

Open the bucket

Upload file (example: index.html)

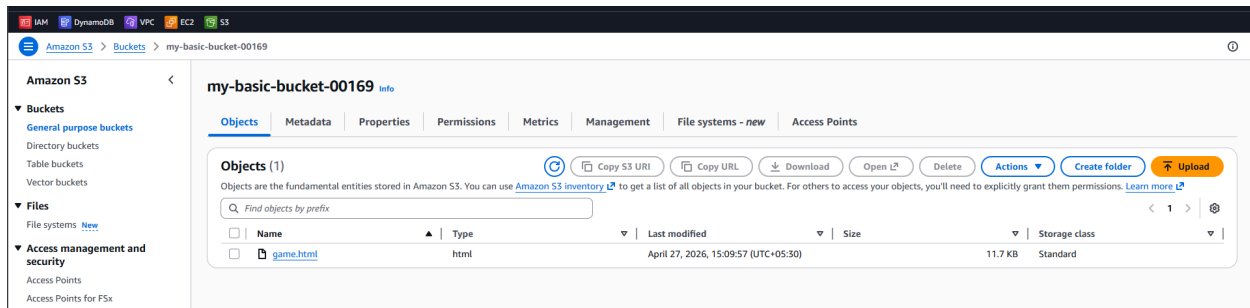
Click file

→

copy Object URL

Make object public (if needed)

Access file in browser



game.html Info [Copy S3 URI](#) [Download](#) [Open L](#)

Properties | Permissions | Versions

Object overview

Owner 4210c6727dd18630d3bc0320a0569eba934ad307124e4582fe658523c111445e	S3 URI s3://my-basic-bucket-00169/game.html
AWS Region US East (N. Virginia) us-east-1	Amazon Resource Name (ARN) arn:aws:s3::my-basic-bucket-00169/game.html
Last modified April 27, 2026, 15:09:57 (UTC+05:30)	Entity tag (Etag) aeac48f83886b427fd3d0fb1b143e23c
Size 11.7 KB	Object URL https://my-basic-bucket-00169.s3.us-east-1.amazonaws.com/game.html
Type html	
Key game.html	

Object management overview

The following bucket properties and object management configurations impact the behavior of this object.

Bucket properties

Bucket Versioning

Management configurations

Replication status

Block public access (bucket settings) [Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access

Off

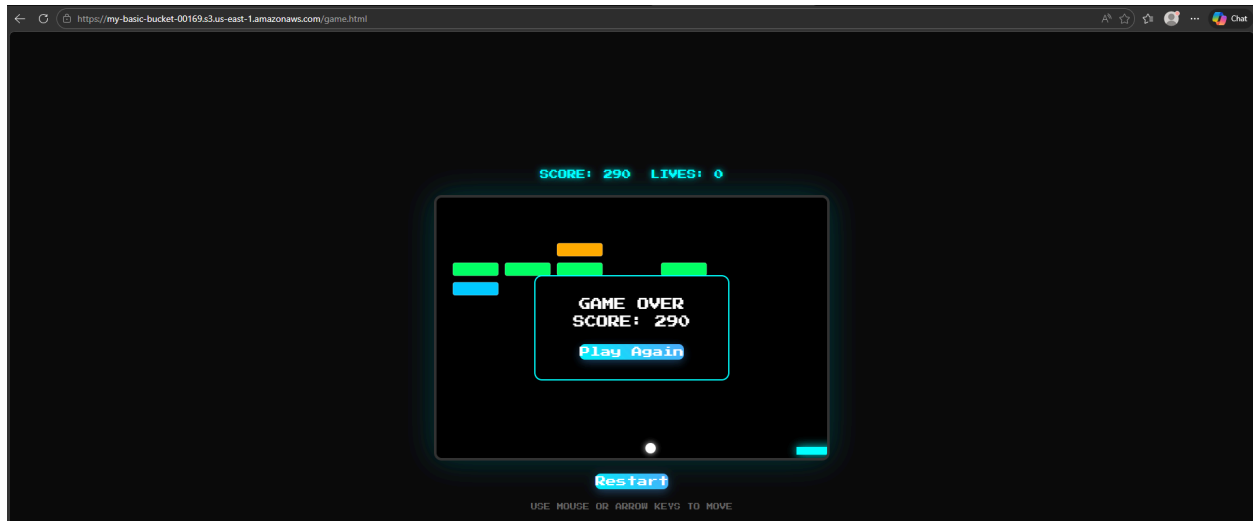
► Individual Block Public Access settings for this bucket

Bucket policy [Edit](#) [Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadOnlyObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3::my-basic-bucket-00169/*"
    }
  ]
}
```

[Copy](#)



Q3. Create VPC with Public Subnet

Build your own network.

Tasks:

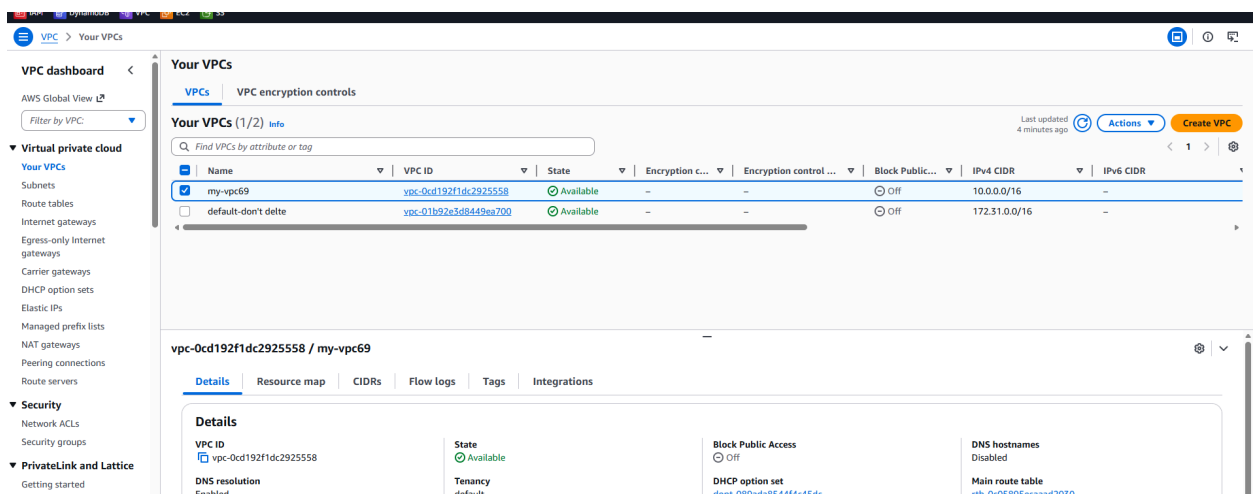
Go to Amazon VPC

→

Create VPC

Name:

my-vpc



CIDR

10.0.0.0/16

vpc-0cd192f1dc2925558 / my-vpc69		
Details Resource map CIDRs Flow logs Tags Integrations		
IPv4 CIDRs Info Edit CIDRs		
Address family	CIDR	Status
IPv4	10.0.0.0/16	Associated

Create Subnet:

Name:

public-subnet

CIDR

10.0.1.0/24

Create Internet Gateway:

Name:

my-igw

Attach to

my-vpc

Create Route Table:

Add route:

0.0.0.0/0 → Internet Gateway

Associate with public subnet

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☒	my-dear-rt69	rtb-04399f0e10b82cfee	subnet-01671dc9c4acb3...	-	No	vpc-0cd192f1dc2925558 my-v...	107970445632
☐	-	rtb-0c05895ecaad2030	-	-	Yes	vpc-0cd192f1dc2925558 my-v...	107970445632
☐	def	rtb-04c0b7f9e47e0b113	-	-	Yes	vpc-01b92e3d8449ea700 def...	107970445632

rtb-04399f0e10b82cfee / my-dear-rt69

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-092b08d0f485a209f	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Q4. Launch EC2 Inside Your VPC

Use your custom network.

Tasks:

Go to EC2

→

Launch Instance

Name:

vpc-ec2

Instances (1/2) Info

Saved filter sets: Choose filter set

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
public	i-036ec4831f534d9ce	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	-	3.231.159.62	-
assign-ec2	i-0d2de7c6fc0c00290	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1c	ec2-98-93-99-176.com...	98.93.99.176	-

i-036ec4831f534d9ce (public)

Details | Status and alarms | Monitoring | Security | **Networking** | Storage | Tags

VPC ID
vpc-0cd192f1dc2925558 (my-vpc69)

Subnet ID
subnet-01671dc9c4acb37e4 (public-subnet)

Availability zone
us-east-1a

Availability zone ID
use1-az1

Outpost ID
-

IP addresses

Public IPv4 address
3.231.159.62 | open address

Private IPv4 addresses
10.0.1.53

IPv6 addresses
-

Secondary private IPv4 addresses
-

Carrier IP addresses (ephemeral)
-

Hostname and DNS

Public DNS
-

Private IP DNS name (IPv4 only)
ip-10-0-1-53.ec2.internal

IPv4-only IP based name: A record only
-

Dualstack - IP based name: A and AAAA record
-

IPv6-only - IP based name: AAAA record only
-

Public hostname type
-

Select VPC
my-vpc

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

vpc-0cd192f1dc2925558 / my-vpc69

Details

VPC ID
vpc-0cd192f1dc2925558

State
Available

DNS resolution
Enabled

Tenancy
default

Main network ACL
acl-0e8316095bac400ce

Default VPC
No

IPv6 CIDR (Network border group)
-

Network Address Usage metrics
Disabled

Encryption control ID
-

Encryption control mode
-

Block Public Access
Off

DHCP option set
dopt-085ada8544f4c45dc

IPv4 CIDR
10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Disabled

Main route table
rtb-0c05895ecaaad2030

IPv6 pool
-

Owner ID
107970445632

Resource map

VPC
Your AWS virtual network

Subnets (1)
Subnets within this VPC

us-east-1a

public-subnet

Route tables (2)
Route network traffic to resources

my-dear-rt69

rtb-0c05895ecaaad2030

Network Connections (1)
Connections to other networks

my-igw

Select Subnet:
public-subnet

subnet-01671dc9c4acb37e4 / public-subnet

Details

Subnet ID subnet-01671dc9c4acb37e4 IPv4 CIDR 10.0.1.0/24 Availability Zone us-east-1 (us-east-1a) Network ACL acl-0e8316095bac400ce Auto-assign customer-owned IPv4 address No IPv6 CIDR reservations - Resource name DNS AAAA record Disabled	Subnet ARN arn:aws:ec2:us-east-1:107970445632:subnet/subnet-01671dc9c4acb37e4 Available IPv4 addresses 250 Network border group us-east-1 Default subnet No Customer-owned IPv4 pool - IPv6-only No DNS64 Disabled	State Available IPv6 CIDR - VPC vpc-0cd192f1dc2925558 my-vpc69 Auto-assign public IPv4 address No Outpost ID - Hostname type IP name Owner 107970445632	Block Public Access Off IPv6 CIDR association ID - Route table rtb-04399f0e10b82cfee my-dear-r169 Auto-assign IPv6 address No IPv4 CIDR reservations - Resource name DNS A record Disabled
--	--	---	---

Flow logs | **Route table** | Network ACL | CIDR reservations | Sharing | Tags

Route table: rtb-04399f0e10b82cfee / my-dear-r169

Routes (2)

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-092b08d0f485a209f

Enable Auto-assign Public IP
 Security Group:
 Allow SSH 22) from your IP
 Allow HTTP 80) from anywhere (0.0.0.0/0)

https://us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-036ec4831f534d9ce&osUser=ubuntu®ion=us-east-1&sshPort=22

0 IAM 0 DynamoDB 0 VPC 0 EC2 0 S3

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

* Documentation: <https://help.ubuntu.com>
 * Management: <https://landscape.canonical.com>
 * Support: <https://ubuntu.com/pro>

System information as of Mon Apr 27 12:53:35 UTC 2026

```
System load: 0.09      Temperature: ~273.1 C
Usage of /: 27.2% of 6.71GB   Processes: 116
Memory usage: 27%      Users logged in: 1
Swap usage: 0%          IPv4 address for ens5: 10.0.1.53
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
 See <https://ubuntu.com/esm> or run: `sudo pro status`

Last login: Mon Apr 27 10:32:50 2026 from 196.1.1.6
 ubuntu@ip-10-0-1-53:~\$

i-036ec4831f534d9ce (public)
 PublicIPs: 3.231.159.62 PrivateIPs: 10.0.1.53

Q5. Attach IAM Role to EC2 and Access S3 Connect EC2 with S3 using role.

Tasks:

Go to AWS IAM

→

Create Role

Use case: EC2

Attach policy:

AmazonS3FullAccess

Roles (11) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

[Delete](#) [Create role](#)

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Service-Linker)	16 minutes ago
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linker)	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linker)	-
☐	fordynamodb-role-1mcb05if	AWS Service: lambda	-
☐	fordynamodb-role-2xbgy2w3	AWS Service: lambda	-
☐	marksfunction-role-l0fivy73	AWS Service: lambda	-
☐	s3-access	AWS Service: ec2	-
☐	sales_total_function-role-v0l5rl0s	AWS Service: lambda	3 days ago
☐	sales_total_function2-role-9hcyq7ty	AWS Service: lambda	3 days ago
☐	studentmarks-role-wys489xx	AWS Service: lambda	3 days ago
☐	studentsresult-role-0B4gwlw5	AWS Service: lambda	-

s3-access [Info](#)

Allows EC2 instances to call AWS services on your behalf.

[Delete](#)

Summary [Edit](#)

Creation date April 28, 2026, 09:20 (UTC+05:30)	ARN arn:aws:iam::107970445632:role/s3-access	Instance profile ARN arn:aws:iam::107970445632:instance-profile/s3-access
Last activity 14 minutes ago	Maximum session duration 1 hour	

[Permissions](#) [Trust relationships](#) [Tags](#) [Last Accessed](#) [Revoke sessions](#)

Permissions policies (1) [Info](#)

You can attach up to 10 managed policies.

[Simulate](#) [Remove](#) [Add permissions](#)

Filter by Type
All types

☐	Policy name	Type	Attached entities
☐	AmazonS3FullAccess	AWS managed	3

Go to EC2

→

Select instance

Actions

→

Security

→

Modify IAM Role

Attach role

Verification:

Connect to EC2 terminal

Run

```
inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
inflating: aws/dist/awscli/customizations/sso/index.html
inflating: aws/dist/awscli/topics/s3-case-insensitivity.rst
inflating: aws/dist/awscli/topics/config-vars.rst
inflating: aws/dist/awscli/topics/return-codes.rst
inflating: aws/dist/awscli/topics/topic-tags.json
inflating: aws/dist/awscli/topics/s3-faq.rst
inflating: aws/dist/awscli/topics/s3-config.rst
inflating: aws/dist/awscli/topics/ddb-expressions.rst
inflating: aws/dist/awscli/data/metadata.json
inflating: aws/dist/awscli/data/ac.index
inflating: aws/dist/awscli/data/cli.json
  creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
buntu@ip-172-31-36-207:~$ sudo ./aws/install
on can now run: /usr/local/bin/aws --version
buntu@ip-172-31-36-207:~$ aws s3 ls
026-04-28 03:54:44 new-buck-s3-fullaa
buntu@ip-172-31-36-207:~$
```

Q6. Create Private Subnet + EC2 (No Internet) Secure backend server.

Tasks:

Create a subnet:

Name:

private-subnet

CIDR:
10.0.2.0/24

Details

Subnet ID

subnet-0c3f03a734d40d2db

IPv4 CIDR

10.0.2.0/24

Availability Zone

use1-az6 (us-east-1d)

Network ACL

acl-00ddfad5506649b11

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-1:107970445632:subnet/subnet-0c3f03a734d40d2db

Available IPv4 addresses

250

Network border group

us-east-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-0402b86e3bee72dd2 | our-vpc

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

107970445632

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-06607da387e13b66c | private-rt

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Route table: rtb-06607da387e13b66c / private-rt

Edit route table association

Routes (1)

Filter routes

Destination

10.0.0.0/16

Target

local

Create Route Table:
Name:
private-rt

Route tables > rtb-06607da387e13b66c

board <

View L2

PC:

vate cloud

s

eways

Internet

ways

1 sets

efix lists

ys

rtb-06607da387e13b66c / private-rt

Actions

Details

Route table ID

rtb-06607da387e13b66c

VPC

vpc-0402b86e3bee72dd2 | our-vpc

Main

No

Owner ID

107970445632

Explicit subnet associations

subnet-0c3f03a734d40d2db / private

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (1)

Find subnet association

Name

private

Subnet ID

subnet-0c3f03a734d40d2db

IPv4 CIDR

10.0.2.0/24

IPv6 CIDR

-

vpc-0402b86e3bee72dd2 / our-vpc Actions ▾

Details [Info](#)

VPC ID
vpc-0402b86e3bee72dd2

DNS resolution
Enabled

Main network ACL
acl-00ddfad506649b11

IPv6 CIDR (Network border group)
-

Encryption control ID
-

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Encryption control mode
-

Block Public Access
Off

DHCP option set
dopt-089ada8544f4c45dc

IPv4 CIDR
10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Disabled

Main route table
rtb-0bc67306419de357d

IPv6 pool
-

Owner ID
107970445632

[Resource map](#) | [CIDRs](#) | [Flow logs](#) | [Tags](#) | [Integrations](#)

Resource map [Info](#) Show all details

VPC
Your AWS virtual network

our-vpc

Subnets (1)
Subnets within this VPC

us-east-1d
private

Route tables (2)
Route network traffic to resources

private-rt
rtb-0bc67306419de357d

Network Connections (0)
Connections to other networks

Launch EC2 Name: private-ec2 Subnet: private-subnet
 Disable public IP
 Security Group: Allow SSH only from VPC CIDR (10.0.0.0/16)

[Instances](#) > [i-0ff3fc078fe417fd](#) > Connect to instance

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

No public IPv4 or IPv6 address assigned
With no public IPv4 or IPv6 address, you can't use EC2 Instance Connect. Alternatively, you can try connecting using [EC2 Instance Connect Endpoint](#).

Instance is not in public subnet
Associated subnet [subnet-0c3f03a734d40d2db \(private\)](#) is not a public subnet. To use EC2 Instance Connect, your instance must be in a public subnet. To make the subnet a public subnet, add a route in the subnet route table to an internet gateway.

Instance ID
i-0ff3fc078fe417fd (private)

Connection type

☒ **Connect using a Public IP**
Connect using a public IPv4 or IPv6 address

☐ **Connect using a Private IP**
Connect using a private IP address and a VPC endpoint

☐ **Public IPv4 address**

☐ **IPv6 address**

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect

Instance summary for i-Off3fc078fe417fdc (private) info		
Updated less than a minute ago		
Instance ID i-Off3fc078fe417fdc	Public IPv4 address -	Private IPv4 addresses 10.0.2.93
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-2-93.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-2-93.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address -	VPC ID vpc-0402b86e3bee72dd2 (our-vpc)	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0c3f03a734d40d2db (private)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:107970445632:instance/i-Off3fc078fe417fdc	
Operator -		

Q7. Create Bastion Host (Jump Server)

Access private EC2 securely.

Tasks:

Launch EC2

Name:

bastion-host

Subnet:

public-subnet

Enable public IP

Security Group Bastion):

Allow SSH 22) from your IP

Modify Private EC2 Security Group:

Allow SSH only from Bastion Security Group (not IP)

VPC dashboard

AWS Global View

Filter by VPC:

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

PrivateLink and Lattice

- Getting started
- Endpoints
- Endpoint services

Your VPCs

VPCs | VPC encryption controls

Last updated less than a minute ago

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	default-don't delete	vpc-01b92e3d8449ea700	Available	-	-	Off	172.31.0.0/16	-
☒	our-vpc	vpc-0402b86e3bee72dd2	Available	-	-	Off	10.0.0.0/16	-

vpc-0402b86e3bee72dd2 / our-vpc

Details | Resource map | CIDRs | Flow logs | Tags | Integrations

Details

VPC ID
vpc-0402b86e3bee72dd2

DNS resolution
Enabled

Main network ACL
acl-00ddfad5506649b11

IPv6 CIDR (Network border group)
-

Encryption control ID
-

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Encryption control mode
-

Block Public Access
Off

DHCP option set
dopt-089ada8544f4c45dc

IPv4 CIDR
10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Disabled

Main route table
rtb-0bc67306419de357d

IPv6 pool
-

Owner ID
107970445632

Route tables (2/4)

Find route tables by attribute or tag

Last updated 2 minutes ago

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☒	private-rt	rtb-06607da387e13b66c	subnet-0c3f03a734d40d...	-	No	vpc-0402b86e3bee72dd2 our...	107970445632
☐	-	rtb-0bc67306419de357d	-	-	Yes	vpc-0402b86e3bee72dd2 our...	107970445632
☐	def	rtb-04c0b7f9e47e0b113	-	-	Yes	vpc-01b92e3d8449ea700 def...	107970445632
☒	public	rtb-0c6599dbf088434ec	subnet-06e9a5bdd59ed3...	-	No	vpc-0402b86e3bee72dd2 our...	107970445632

Route tables: rtb-06607da387e13b66c, rtb-0c6599dbf088434ec

Subnets (2/8)

Find subnets by attribute or tag

Last updated 3 minutes ago

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6
☒	public	subnet-06e9a5bdd59ed373a	Available	vpc-0402b86e3bee72dd2 our...	Off	10.0.1.0/24	-	-
☐	def5	subnet-08a1f91e338596d3e	Available	vpc-01b92e3d8449ea700 def...	Off	172.31.64.0/20	-	-
☐	def6	subnet-015c59854a0c5d2d8	Available	vpc-01b92e3d8449ea700 def...	Off	172.31.32.0/20	-	-
☒	private	subnet-0c3f03a734d40d2db	Available	vpc-0402b86e3bee72dd2 our...	Off	10.0.2.0/24	-	-

Subnets: subnet-06e9a5bdd59ed373a, subnet-0c3f03a734d40d2db

Instances (1/2) [Info](#)

[Connect](#)
[Instance state](#)
[Actions](#)
[Launch instances](#)

Saved filter sets
 Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state: [running](#) [Clear filters](#)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
pvt	i-0427159da43f2dff0	Running	t3.micro	3/3 checks passed	View alarms	us-east-1d	-	-	-
Bastion-Host	i-063ee97fbb10a5ff6	Running	t3.micro	3/3 checks passed	View alarms	us-east-1d	-	54.162.120.74	-

i-063ee97fbb10a5ff6 (public)

[Details](#)
[Status and alarms](#)
[Monitoring](#)
[Security](#)
[Networking](#)
[Storage](#)
[Tags](#)

▼ **Instance summary** [Info](#)

Instance ID i-063ee97fbb10a5ff6	Public IPv4 address 54.162.120.74 open address	Private IPv4 addresses 10.0.1.218
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-1-218.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-1-218.ec2.internal	

Q8. Verify Private Access via Bastion

Tasks:

Connect to Bastion host

From Bastion:

```
ssh -i key.pem ubuntu@<private-ec2-private-ip>
```

Observation:

Private EC2 has no public IP

Access only through Bastion

```

ubuntu@ip-10-0-1-218:~$ chmod 400 first-ec2.pem
ubuntu@ip-10-0-1-218:~$ ssh -i "first-ec2.pem" ubuntu@10.0.2.246
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Apr 28 13:33:47 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   26.3% of 6.71GB Processes:      113
Memory usage: 25%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.2.246

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;

```

Q9. Create Two IAM Users (Access vs No Access)

e

Test permission control.

Tasks:

Create User1

Name:

s3-user-allowed

Enable console access

Attach policy:

AmazonS3FullAccess

full-access Info

Delete

Summary

ARN

arn:aws:iam::107970445632:user/full-access

Created

April 27, 2026, 16:19 (UTC+05:30)

Console access

Enabled without MFA

Last console sign-in

Today

Access key 1

Create access key

Permissions

Groups

Tags

Security credentials

Last Accessed

Permissions policies (2)

Remove

Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

All types

Search

< 1 >

☐	Policy name	Type	Attached via
☐	AmazonS3FullAccess	AWS managed	Directly
☐	IAMUserChangePassword	AWS managed	Directly

Permissions boundary (not set)

Create User2

Name:

s3-user-denied

Enable console access

Do not attach any policy

Summary

ARN

arn:aws:iam::107970445632:user/no-access

Created

April 27, 2026, 16:20 (UTC+05:30)

Console access

Enabled without MFA

Last console sign-in

Today

Access key 1

Create access key

Permissions

Groups

Tags

Security credentials

Last Accessed

Permissions policies (1)

Remove

Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

All types

Search

< 1 >

☐	Policy name	Type	Attached via
☐	IAMUserChangePassword	AWS managed	Directly

Permissions boundary (not set)

Generate policy based on CloudTrail events

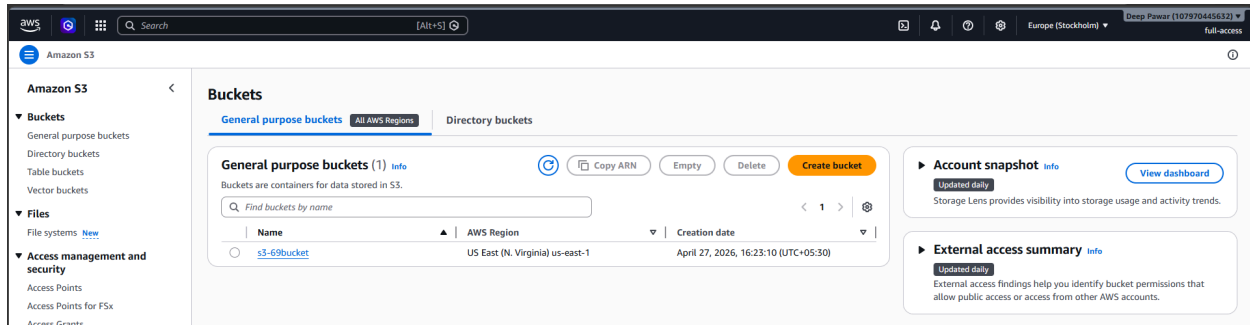
You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

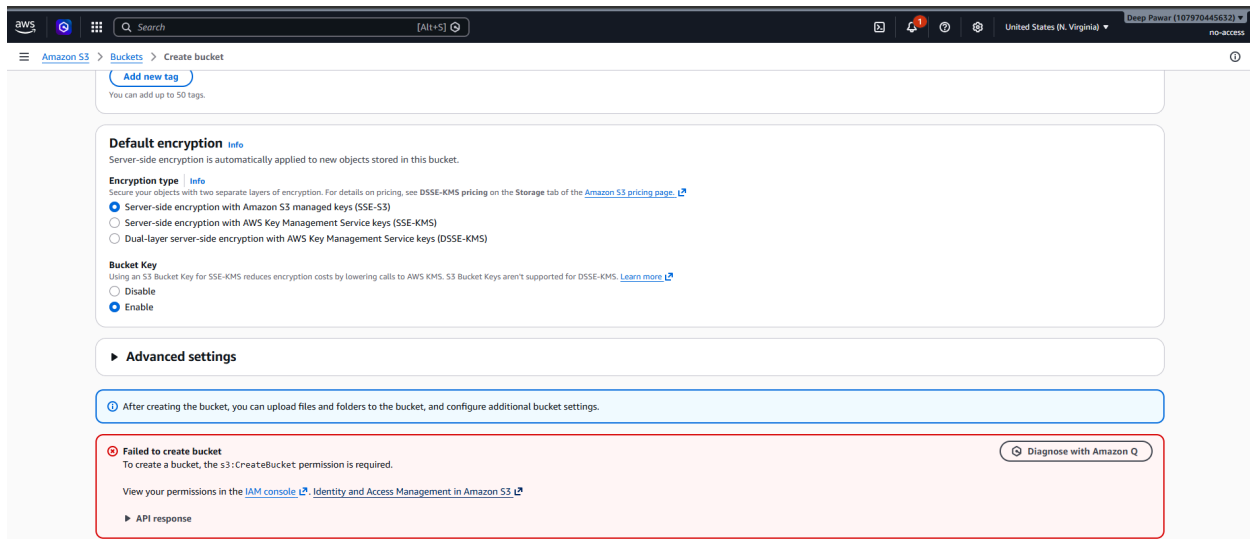
Q10. Test S3 Access (Permission Check)

Tasks:
Login as User1
Go to S3
Upload file
Should succeed



Login as User2
Go to S3
Try to upload file
Should fail Access Denied)

• +



lambda_function.py

```
1 # Create a AWS Lambda function that takes two numbers as input using a test event and return
2
3 import json
4
5 def lambda_handler(event, context):
6     # Extract the two numbers from the event
7     num1 = event['num1']
8     num2 = event['num2']
9
10    # Add the two numbers
11    result = int(num1) + int(num2)
12
13    print(f"The sum of {num1} and {num2} is {result}")
14
15    # Return the result
16    return {
17        'statusCode': 200,
18        'body': json.dumps({
19            'result': result
20        })
21    }
```

Create new test event

☒ Private

This event is only available in the Lambda Console and to the event creator. You can configure a total of ten. [Learn more](#)

☐ Shareable

This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional

test2

Event JSON

```
1 {
2   "num1": "30",
3   "num2": "50"
4 }
```

PROBLEMS

OUTPUT

CODE REFERENCE LOG

TERMINAL

Execution Results

Response:

```
{
  "statusCode": 200,
  "body": "{\"result\":80}"
}
```

The area below shows the last 4 KB of the execution log.